

[C272/SQP362]

Physics
Higher
Specimen Marking Instructions
for use in and after 2012

NATIONAL
QUALIFICATIONS

Physics Higher

Specimen Question Paper

Marking scheme

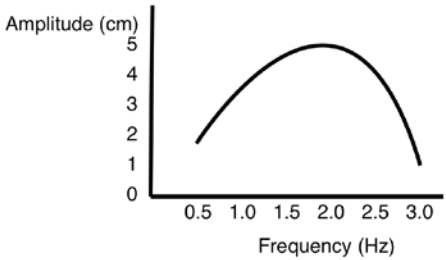
Section A

1.	B	11.	A
2.	A	12.	A
3.	C	13.	B
4.	D	14.	A
5.	B	15.	C
6.	D	16.	D
7.	B	17.	E
8.	A	18.	A
9.	B	19.	A
10.	C	20.	C

Physics – Higher Specimen Question Paper						
Sample Answer and Mark Allocation				Notes	Inner Margin	Outer Margin
21	(a)	(i)	Initial horizontal component = $v \cos \theta$ $= 50 \times \cos 35$ $= 41 \text{ ms}^{-1}$ 1	no formula half	1	9
		(ii)	Initial vertical component = $v \sin \theta$ $= 50 \times \sin 35$ $= 29 \text{ ms}^{-1}$ 1	no formula half	1	
	(b)		$v^2 = u^2 + 2as$ $0 = 29^2 + 2 \times (-9.8) \times s$ $s = 29^2 / 19.6$ $= 43 \text{ m}$ 1/2 1/2 1		2	
	(c)	(i)	$v = u + at$ $v = 29 - 9.8 t$ $t = 0 - 29 / -9.8$ $= 2.96 \text{ (s)}$ 1/2 1/2 1/2 Total time (up and down) = 2.96×2 $= 5.92 \text{ (s)}$ 1/2 $s_h = v_h \times t$ $= 41 \times 5.92$ $(= 240 \text{ m})$ 1/2 1/2		3	
		(ii)	Reduced maximum height 1 Smaller horizontal distance travelled 1	No requirement to show position of max height, nor unsymmetrical trajectory	2	
22			Demonstrates no understanding 1 Limited understanding 2 Reasonable understanding 3 Good understanding 3	Open ended question	3	3
23	(a)		$v = H_0 d$ $H_0 = 2.3 \times 10^{-18} \text{ (s}^{-1}\text{)}$ $5.5 \times 10^5 = 2.3 \times 10^{-18} \times d$ $d = 2.4 \times 10^{23} \text{ m}$ 1/2 1 1/2 1	Look up Hubble's constant	3	4
23	(b)		Wavelength (or frequency) 1	Single word answer is acceptable	1	
24			Reference to $F = ma$ (or equivalent) F and m increase by same amount hence a is constant 1 1		2	2

25	(a)	Volume = $\frac{4}{3} \pi r^3$ = $\frac{4}{3} \pi (6.05 \times 10^6)^3$ = $9.27 \times 10^{20} \text{ m}^3$ $\rho = m/V$ $5.24 \times 10^3 = m / 9.27 \times 10^{20}$ $m = 4.86 \times 10^{24} \text{ (kg)}$ Spacecraft orbits Venus	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	Selected from data booklet No formula half	3	6
	(b)	Period ² Radius ³ Period ² /Radius ³ 4.84 512 9.45 16.3 1728 9.43 38.7 4096 9.45 Calculation of T ² and R ³				

	(d)	Flow of molecules/air current	1	or similar statement, indicating bulk movement	2	
		Energy/heat removed from chip	1			
28	(a)	Blue light has higher frequency/energy per photon than red light.	1	Or similar statement comparing blue and red light	2	5
		Photons of red light do not have enough energy to eject electrons	1	Or similar statement in terms of threshold frequency or work function		
	(b)	$E_k = hf - hf_0$ $= 6.63 \times 10^{-34} \times 7.0 \times 10^{14} - 2.0 \times 10^{-19}$ $= 2.461 \times 10^{-19} \text{ J}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1	Half mark for look up of h Half mark for calculation of hf Half mark for subtraction of work function	3	
29	(a)	(i)	$d = 1/200 \times 1000 = 5 \times 10^{-6} \text{ (m)}$ $n \lambda = d \sin \theta$ $2 \times \lambda = 5 \times 10^{-6} \times \sin 11.8$ $\lambda = 4.77 \times 10^{-6} \text{ m}$	1 $\frac{1}{2}$ $\frac{1}{2}$ 1		8
		(ii)	Colour is green	1	Look up spectral line in data sheet	
	(b)	Maxima are spaced further apart d and $\sin \theta$ are inversely proportional	1 1		2	
	(c)	Dark lines are absorption lines caused by certain elements in sun absorbing particular frequencies	1 1		2	
30	(a)	$R = V/I$ $= 12/30 \times 10^{-6}$ $= 400\,000 \, \Omega$	$\frac{1}{2}$ $\frac{1}{2}$ 1	Or equivalent in k Ω , M Ω	2	9
	(b)	As voltage across the capacitor increases, voltage across the resistor increases. To maintain the same current through the resistor, the resistance must be decreased.	1 1	No marks for a statement with no justification. No marks for correctly stating resistance decrease, with an incorrect justification.	2	

(c)	(i)	$Q = It$ $= 30 \times 10^{-6} \times 30$ $= 900 \times 10^{-6} \text{ C}$	$\frac{1}{2}$ $\frac{1}{2}$ 1	Or equivalent milli/micro	2	
	(ii)	$C = Q/V$ $200 \times 10^{-6} = 900 \times 10^{-6} / V$ $V = 4.5 \text{ V}$ Therefore voltage across resistor is $12 - 4.5 = 7.5 \text{ V}$	$\frac{1}{2}$ $\frac{1}{2}$ 1 1		3	
31		Demonstrates no understanding Limited understanding Reasonable understanding Good understanding	0 1 2 3	Open ended question	3	3
32	(a)	 Peak in approx correct position Ends of graph in approx correct position	1 1	Indication of axes scales required	2	7
	(b)	Amplitude of peak has a peak frequency at or about 2 Hz	1		1	
	(c)	Repeat measurements More measurements, especially close to and on either side of the peak	1 1	Or any other valid improvement	2	
	(d)	Any valid investigation Examples include: Finding the relationship between the frequency that results in the peak amplitude and height of a building (i.e. resonant frequency vs height) Investigating how to reduce the amplitude of vibration (i.e structural improvements) Brief plan of how investigation would be undertaken	1 1		2	

[END OF SPECIMEN MARKING INSTRUCTIONS]